CHEMICAL EQUILIBRJUM

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→ Forward Reaction

→ Backward Reaction

→ Irreversible Reaction

→ Reversible Reaction

→ Types of chemical equilibrium

- → Homogeneous
- → Heterogenous

→ Characteristics of chemical equilibrium

→ Law of mass action and equilibrium constant.

→ Relation between K_p and K_c

→ La Chatalier Principle.

Forward Reaction:- ⟶

A forward reaction is which products are produced from reatants and it goes from left to right in a reversible reaction.

Example:- $N_2 + 3H_2 \longrightarrow 2NH_3$

i.e reactants ⟶ products.

Backward Reaction:- ⟵

A backward reaction is a reaction in which reactants are produced from products and it goes from right to left in a reversible reaction.

i.e, Reactants ⟵ products.

Example: $2NH_3 \longrightarrow N_2 + 3H_2$

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Homogenous Equilibrium:-

The chemical equilibrium in which all reactants and products are in the same physical state is called homogenous equilibrium

Example:-

1. $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

2. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

All products and reactants are in same gaseous state

3. $CH_3COOH_{(l)} + C_2H_5OH_{(l)} \rightleftharpoons CH_3COOC_2H_{5(l)} + H_2O_{(l)}$

All products and reactants are in liquid state.

Heterogeneous Equilibrium:-

A chemical equilibrium in which at least one product or reactant is in different physical state (or phase) than the others, is called a heterogeneous equilibrium.

Example:-

1. $CaCO_{3(s)} \overset{\Delta}{\rightleftharpoons} CaO_{(s)} + CO_{2(g)}$

~~The~~ Two components are in solid state and one component in gaseous state.

2. ~~Fe~~ $3Fe_{(s)} + 4H_2O_{(g)} \rightleftharpoons Fe_3O_{4(s)} + 4H_{2(g)}$

One product and one reactant in solid state but one reactant and one product in gaseous state.

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Characteristics of Chemical Equilibrium:-

There some characteristics of chemical Equilibrium. They are:-

1. Reversibility of reaction: Chemical equilibrium is related to only to reversible reactions. It is not applicable to irreversible reaction.

2. Stability of equilibrium:- If the external conditions (Pressure, temperature, and concentration) are not changed, the system will stay their forever.

3. Easy approachability from both sides: The equilibrium can be attained from both directions.

4. Incompleteness of reaction: A reversible reaction is never complete, because due to backward reaction, the initial reactant are always ~~report~~ reproduced.

5. Ineffectiveness of catalysts: The catalysts have no effect in chemical equilibrium. Catalysts speed up both the forward and backward reaction to the same degree.

6. Closed System:- It is only attained in closed system. So that part of any product or reactant cannot escape out.

7. Dynamic Nature:- Chemical equilibrium is dynamic. Although apparently reaction stops when chemical equilibrium is attained.

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But in reality the forward and backward reactions continue to occur but since their rates are equal & no change is visible.

⊟ Law of mass action and equilibrium constant:-

In 1867 mathematician C.M Guldberg and Chemist P. Waage of Norway invented a law regarding the rate of chemical reactions, which became known as Law of mass action.

The law is as follows:-

At a constant temperature the rate of a chemical reaction at any instant is directly proportional to the active mass (i.e molar concentration or partial Pressure) of the reactants at that instant participitating in the reaction.

⊟ Le chatelier Principle:-

If a change occurs in one of the controlling factors, such as temperature, pressure, concentration etc. under which a system is in equlibriom the system will tend to adjust itself in such a way so as to reduce the effect of that change.

There are three ways in which the stress can be caused on a chemical equilibrium:

1. Changing the concentration of reactant or product.
2. Changing the pressure (or volume) of the system.
3. Changing the temperature.

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If a change in concentration, pressure or temperature is caused to a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimize the change.

Effect of change in concentration:-

When concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

When N_2 or H_2 is added to the equilibrium stage, the equilibrium will shift to the right side so ~~that~~ as to reduce the concentration of the N_2/H_2.

To have a better yield of NH_3, One of the reactants should be added in excess.

Effect of change of temperature:-

When temperature of a reaction is increased the equilibrium shifts in a direction in which heat is absorbed.

Ammonia synthesis reaction is a exothermic reaction.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -92kJ$$

As heat is increased in this reaction, an increase of reaction will cause the equlibrium to shift to the left and NH_3 production will be decreased.

Exothermic Reaction

$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

Effect of change of Pressure:-

When pressure is increased on a gaseous equilibrium reaction, the equilibrium will shift in a direction which tends to decrease the pressure.

▣ Equilibrium Constant:- [সাম্য ধ্রুবক]

Let us consider following reaction,

$$A + B \rightleftharpoons C + D$$

Reactants   Products

According to the law of mass action, the rate of forward reaction,

$R_f \propto [A][B]$

$R_f = K_1[A][B]$

Where,
$[A]$ = Molar concentration of the reactant A
$[B]$ = Molar concentration of the reactant B.
K_1 = Rate constant of the forward reaction

Similarly the rate of Backward reaction,

$R_b \propto [C][D]$

$R_b = K_2[C][D]$

Where,
$[C]$ = Molar concentration of the product C
$[D]$ = Molar concentration of the product D
K_2 = Rate constant of the backward reaction.

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At the equilibrium, the rate of forward and backward reaction will be same,

$$R_f = R_b$$

$$\Rightarrow K_1[A][B] = K_2[C][D]$$

$$\Rightarrow \frac{[A][B]}{[C][D]} = \frac{K_2}{K_1}$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C][D]}{[A][B]}$$

$$\Rightarrow K_c = \frac{[C][D]}{[A][B]}$$

$\therefore K_c$ is called the molar equilibrium constant or simply equilibrium constant of the reaction.

$K_c = \frac{[C][D]}{[A][B]}$ is the mathematical expression for the law of chemical equilibrium.

$$aA + bB \rightleftharpoons cC + dD$$

Forward reaction,

$$R_f \propto [A]^a [B]^b$$

$$R_f = K_1 [A]^a [B]^b$$

Backward reaction,

$$R_b \propto [C]^c [D]^d$$

$$R_b = K_2 [C]^c [D]^d$$

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At equilibruim,

$$R_f = R_b$$

$$\Rightarrow K_1 \cdot \frac{[A]^a [B]^b}{[C]^c [D]^d} = K_2 [C]^c [D]^d$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\Rightarrow K_c = \frac{C_C^c \times C_D^d}{C_A^a \times C_B^b}$$

$$\Rightarrow K_p = \frac{P_C^c \times P_D^d}{P_A^a \; P_B^b}$$

☐ Relation between K_p and K_c:-

Let us consider a general reversible gaseous reaction.

$$aA + bB + cC \rightleftharpoons dD + eE + fF \quad \text{——} (1)$$

Rate of forward reaction,

$$R_f \propto [A]^a [B]^b [C]^c$$

$$R_f = K_1 [A]^a [B]^b [C]^c \quad \text{——} (2)$$

Rate of backward reaction,

$$R_b \propto [D]^d [E]^e [F]^f$$

$$R_b = K_2 [D]^d [E]^e [F]^f \quad \text{——} (3)$$

At equilibrium,

$$R_f = R_b$$

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$$\Rightarrow K_1 [A]^a [B]^b [C]^c = K_2 [D]^d [E]^e [F]^f$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c}$$

$$\Rightarrow K_c = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c} \text{ ——— } (4)$$

$$\Rightarrow K_c = \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \text{ ——— } (5)$$

$$\Rightarrow K_p = \frac{P_D^d \times P_E^e \times P_F^f}{P_A^a \times P_B^b \times P_C^c} \text{ ——— } (6)$$

We know,

$$PV = nRT$$

$$\Rightarrow P = \frac{n}{V} RT \text{ ——— } 7$$

$$\Rightarrow P = CRT \text{ ——— } (7)$$

Putting the value of P in equation — (6)

$$K_p = \frac{(CRT)_D^d (CRT)_E^e (CRT)_F^f}{(CRT)_A^a (CRT)_B^b (CRT)_C^c}$$

$$= \frac{C_D^d \times (RT)^d \times C_E^e \times (RT)^e \times C_F^f \times (RT)^f}{C_A^a \times (RT)^a \times C_B^b \times (RT)^b \times C_C^c \times (RT)^c}$$

$$= \frac{C_D^d \times C_E^e \times C_F^f (RT)^{d+e+f}}{C_A^a \times C_B^b \times C_C^c \times (RT)^{a+b+c}}$$

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$$= \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \times (RT)^{(d+e+f)-(a+b+c)}$$

$$= K_c RT^{(n_2-n_1)} \quad [\text{from } 5]$$

$$= K_c RT^{\Delta n}$$

Establish relation between K_p and K_c for the given reaction.

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Math:-

1. Some nitrogen and hydrogen gases are pumped into an empty five liter glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2, 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of Kc for the reaction.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

Here,

$$[N_2] = \frac{3.00\,mole}{5.00\,L} = 0.6M$$

$$[H_2] = \frac{2.10\,mole}{5.00L} = 0.42M$$

$$[NH_3] = \frac{0.298\,mole}{5.00} = 0.0596M.$$

$$\therefore K_c = \frac{[NH_3]^2}{[N_2] \times [H_2]^3}$$

$$= \frac{(0.0596)^2}{0.6 \times (0.42)^3}$$

$$= 0.799.$$

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2. The value of K_p at 25°C for the reaction

$$2NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2NOCl_{(g)}$$

is 1.9×10^3 atm^{-1}. Calculate the value of K_c at the same temperature

Here,

$K_p = 1.9 \times 10^3$

$T = 273 + 25 = 298K$

$\cancel{\Delta n}$

$\Delta n = 2 - (2+1) = -1.$

$R = 0.0821$

We know,

$$K_p = K_c (RT)^{\Delta n}$$

$$\Rightarrow K_c = \frac{K_p}{(RT)^{\Delta n}}$$

$$= \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}}$$

$$= 4.6 \times 10^4.$$

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3. The following data was obtained on hydrolysis of methyle acetate at 25°C in 5N hydrochloric acid. Establish it is a first order reaction.

t (secs)	0	4500	7140	∞
ml alkyl used	24.36	29.32	31.72	47.15.

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4. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.

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5. Calculate the vapour pressure lowering caused by the addition of 100g of sucrose (mol mass = 342) to 1000g of water if the ~~water~~ vapour pressure of pure water at 25°c is 23.8 mm Hg.

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